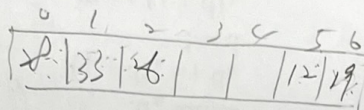


9. A hash table has 7 slots, indexed 0 to 6. It uses the hash function $h(k) = k \bmod 7$ and resolves collisions by linear probing, i.e., if a slot is occupied, the table tries $(\text{index} + 1) \bmod 7$, $(\text{index} + 2) \bmod 7$, and so on, until an empty slot is found. Starting from an empty table, the keys 12, 19, 28, 33, 26 are inserted in this order. In which slot is the key 26 finally stored?

- (A) 5
(B) 2
(C) 6
(D) 0
(E) 1

$$\begin{aligned} 12 \% 7 &= 5 \\ 19 \% 7 &= 5 \\ 28 \% 7 &= 0 \\ 33 \% 7 &= 5 \\ 26 \% 7 &= 5 \end{aligned}$$



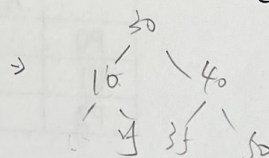
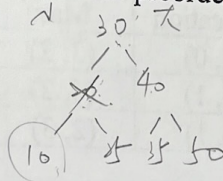
10. A data set is already almost sorted in ascending order, except that only a few adjacent elements are out of order. If we want to use a simple comparison-based sorting algorithm that usually performs close to linear time in such an "almost sorted" case, which algorithm is most suitable?

- (A) Insertion Sort
(B) Heap Sort
(C) Selection Sort
(D) Quick Sort when the worst pivot is chosen every time
(E) Counting Sort for arbitrary integer data with an unbounded range

11. Insert the following numbers into a binary search tree in the given order: 30, 20, 40, 10, 25, 35, 50

Assume that when a node with two children is deleted, it is replaced by its in-order successor. After deleting node 20, what is the preorder traversal of the resulting BST?

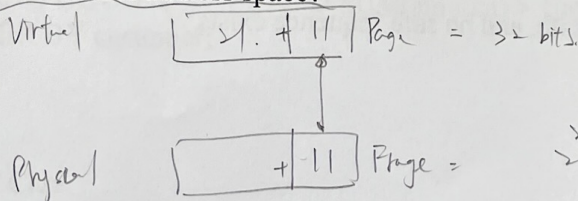
- (A) 30, 10, 25, 40, 35, 50
(B) 25, 10, 30, 35, 40, 50
(C) 30, 25, 40, 10, 35, 50
(D) 30, 25, 10, 40, 35, 50
(E) 25, 30, 10, 40, 35, 50



$\Rightarrow \text{preorder} = 30, 10, 25, 40, 35, 50$

12. A system uses 32-bit virtual addresses, byte-addressable memory, and 8 KB pages. Each page-table entry occupies 4 bytes. What is the size of a single-level page table covering the entire virtual address space?

- (A) 512 KB
(B) 1 MB
(C) 2 MB
(D) 4 MB
(E) 8 MB



$$\frac{8KB}{4B} = 2^{11} = 1167$$

$$2^{11} = 2MB$$

$$13 + 2 = 15$$

注意:背面有試題